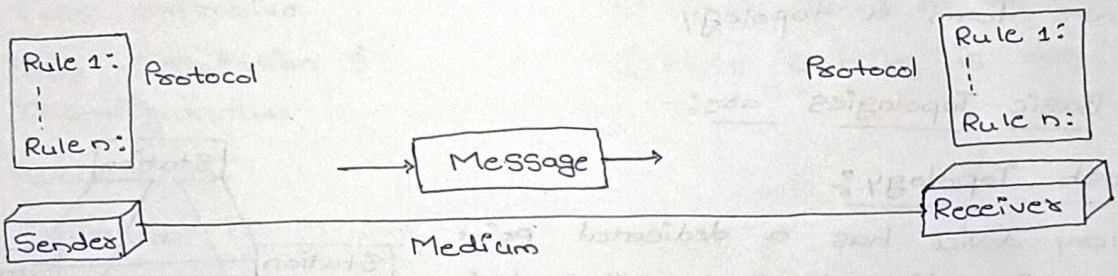
Module - 1

* Data Communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.

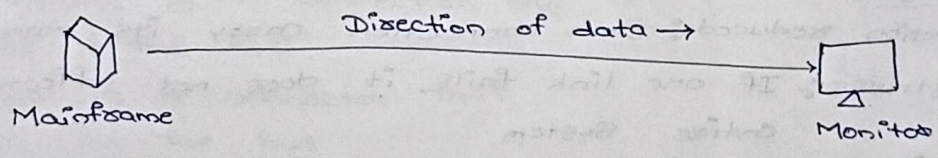
* **Components of a data Communication**

1) Message - Information to be communicated
2) Sender - Send the data message
3) Receiver - Receives the message
4) Transmission Medium - Physical path by which message travels from Sender to receiver.
5) Protocol - Set of rules govern the data Communications

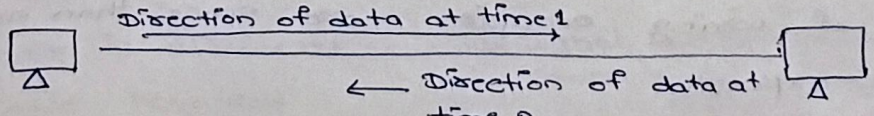


* **Data Flow** :- Communication between two devices

1) **Simplex** :-
   - → The Communication is unidirectional.
   - → only one of the two devices on a link can transmit, the other can only receive.
   - → Eg :- Keyboards



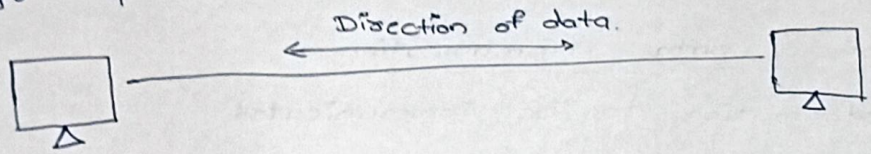
2) **Half-Duplex** :-
   - → In half-duplex mode, each station can both transmit and receive, but not at the same time.
   - → Eg :- Walkie-talkies



3) Full - Duplex :-

→ In full-duplex mode, both stations can transmit & receive simultaneously.

→ Eg :- Telephone



* Physical Topology :-

→ The term physical topology refers to the way in which a network is laid out physically.

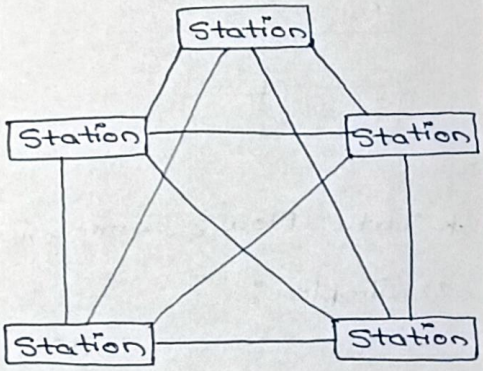
→ Two or more devices connect to a link; two or more links form a topology.

→ 4 Basic Topologies are :-

1) Mesh Topology :-

→ Every device has a dedicated point to point link to every other device

→ For 'n' nodes
  - n(n-1) physical links
  - n(n-1)/2 duplex mode links



→ Eg :- Connection of Telephone in regional offices

* Advantages :-

① Congestion reduced :- Each connection carry its own data load

② Robustness :- If one link fails, it does not affect the entire system

③ Security

④ Easy fault identification.

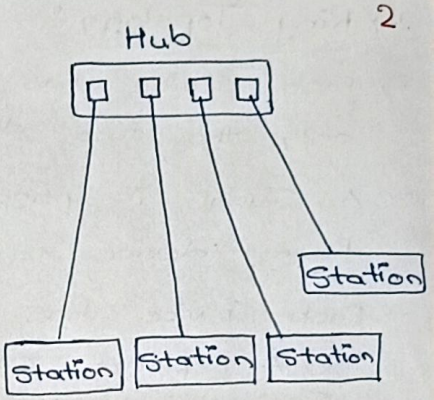
* Disadvantages :-

① Installation & reconnection are difficult.

② Bulk of wiring occupies more space than available space

③ Very expensive.

2) Star Topology :-



→ Point to point connection

→ All the devices are connected to a central controller called a hub

→ Dedicated point-to-point link b/w a device & hub.

→ There is no direct traffic between devices

→ Star topology is used in local area networks (LAN)

* Advantages :-

① Less expensive

② Easy Installation & reconfiguration

③ Robustness

④ Centralized Management

* Disadvantages :-

① Single point of failure (Hub)

② More cabling is required.

3) Bus Topology :-

→ Multipoint Connection

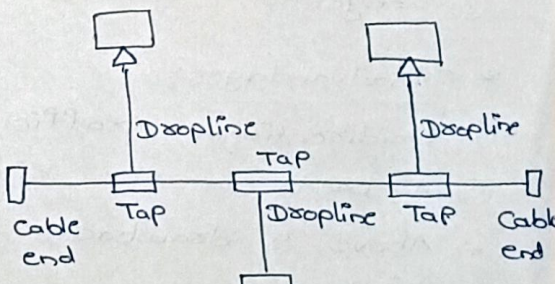
→ All the devices are connected to the single cable called Bus

→ Devices connected to the bus by drop-lines & taps.

→ A dropline is a connection b/w device & the bus

→ A tap is a connector that links to the bus

→ Limit on the number of taps



* Advantages :-

① Easy Installation

② Costs less

③ Redundancy is eliminated

④ Less Cable required

* Disadvantages :-

① Difficult reconnection & fault detection.

② Signal reflection at taps can cause degradation in quality.

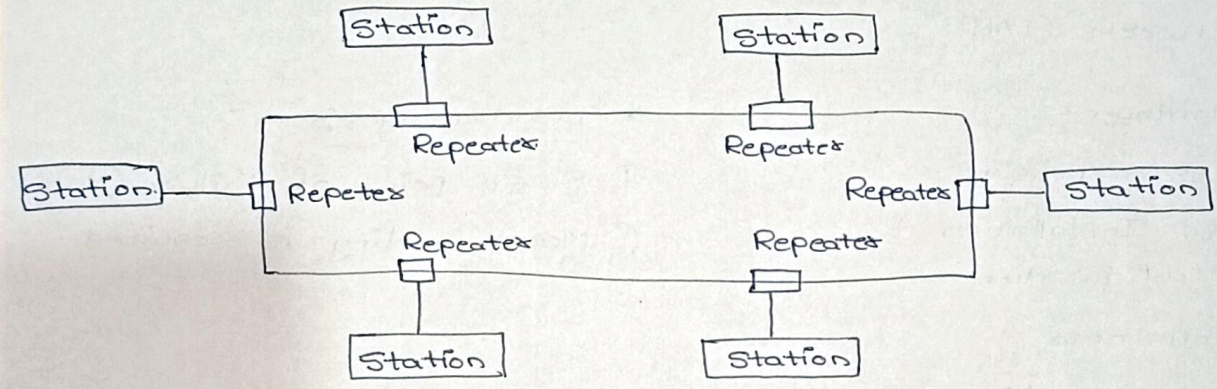
③ A fault in bus cable stops all transmission.

4) Ring Topology :

→ Each device has a dedicated point-to-point connection with only the two devices on either side of it.

→ A signal is passed along the ring in one direction, from device to device, until it reaches its destination.

→ Each device has a repeater
  ↳ A repeater receives a signal on transmission-medium
  ↳ A regenerates & passes the signal to next device.



* Advantages :

① Easy Installation & reconfiguration.

② Fault isolation is simplified.

③ Congestion reduced.

* Disadvantages :

① Unidirectional traffic

② A break in the ring can disable the entire network.

→ Above 2 drawbacks can be overcome by using dual ring.

* Network Types :-

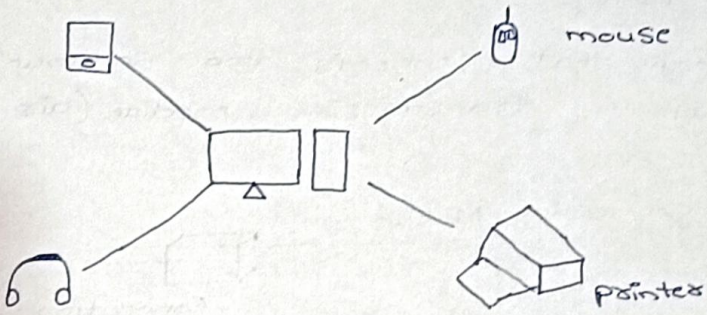
1) Personal Area Networks :- (PAN)

→ In enables short range communication between devices around a person

→ This system makes wireless connection simple, efficient

→ A common example is a computer connecting wirelessly to peripherals like a mouse, keyboard & printer.

→ Eg :- Bluetooth



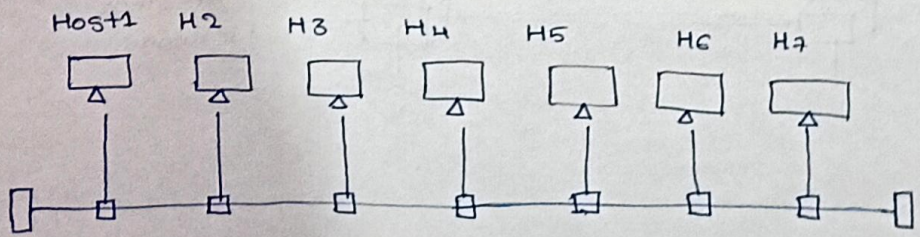
2) Local Area Networks :- (LAN)

→ A LAN is usually privately owned and connects some hosts in a single office, building, or campus.

→ Each host in a LAN has an address that uniquely defines the host in the LAN.

→ A packet sent by a host to another host carries both source host's and destination host's addresses

→ Privately owned Network.



→ Short distances

→ provide local interconnectivity.

3) Wide Area Networks :- (WAN)

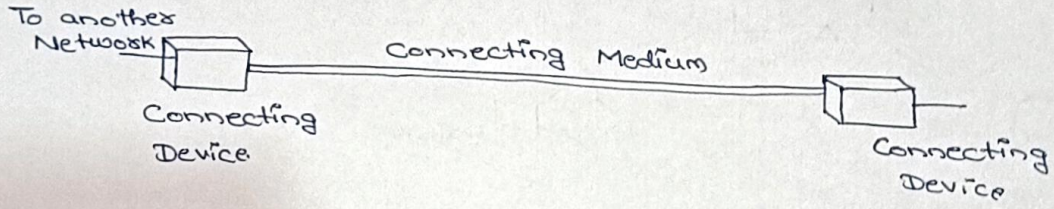
→ A WAN is also an interconnection of devices capable of communication.

→ WAN can cover wider geographical area. It can cover cities, states, countries and even world.

→ Normally, WAN is
  - → created & run by communication companies (Ex: BSNL, Airtel)
  - → leased by an organization that uses it.

* point-to-point WAN :-

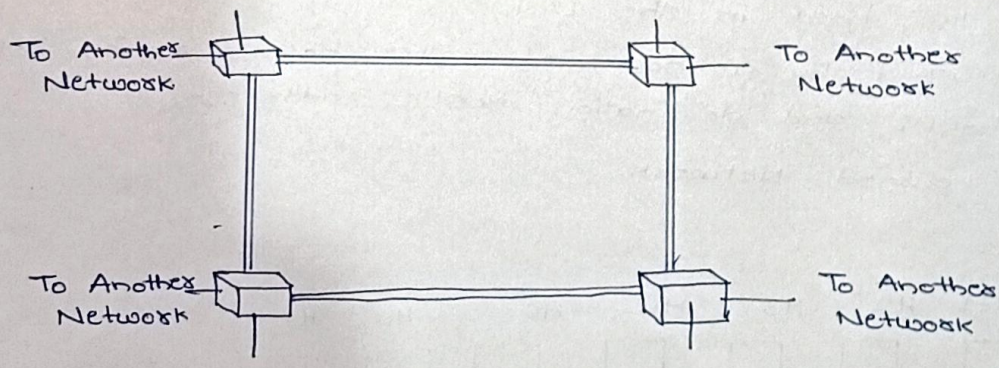
→ It is a network that connects two communicating devices through a transmission media (air or cable).



* Switched WAN :-

→ It is a network with more than two ends.

→ It is a combination of several point-to-point WANs that are connected by switches.



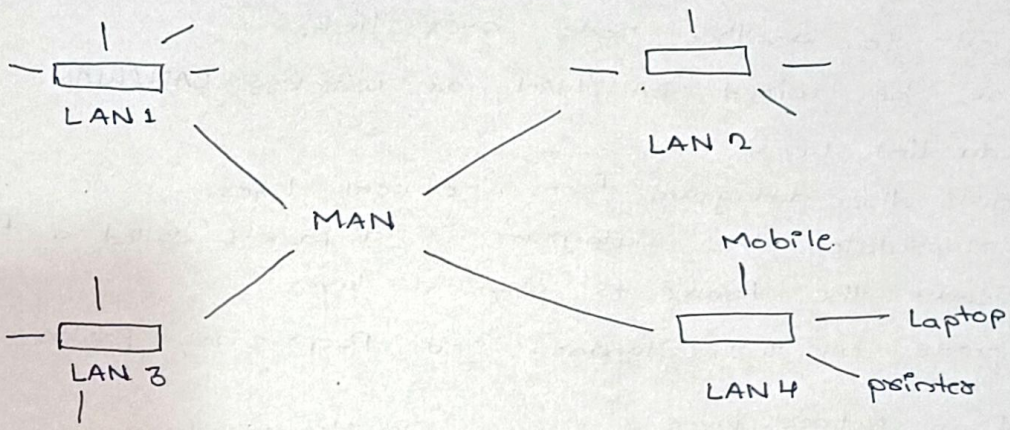
4) Metropolitan Area Networks :- (MAN)

→ Covers a city or large geographical area, larger than a LAN, but smaller than WAN.

→ Connects multiple LANs within a city, providing faster communication & data sharing.

→ MAN Networks provide high reliability because the error rate in this network is very less.

→ Eg :- Cable Television Networks



* TCP / IP Protocol Suite :-

→ Transmission Control Protocol / Internet Protocol.

→ TCP/IP is a protocol suite used in Internet today.

→ It is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality.

→ TCP/IP is thought of as a five-layer model.

1) Physical Layer :-

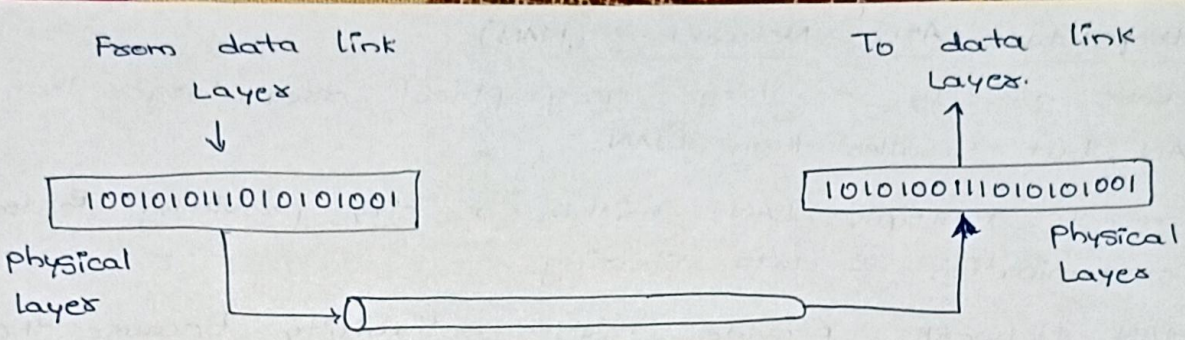
→ Physical layer is responsible for carrying individual bits in a frame across link.

→ Transmission media is another hidden layer under the physical layer

→ Two devices are connected by a transmission medium (cable or wire)

→ The transmission medium does not carry bits; it carries electrical or optical signals.

→ The physical layer receives bits from the data-link layer & sends through the transmission media.



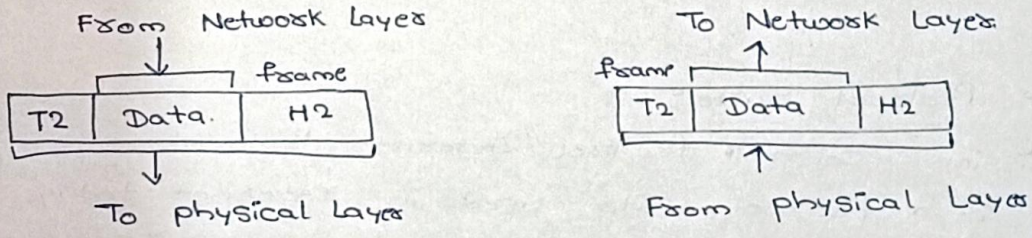
2) Data-Link Layer :-

→ Data Link layer is responsible for moving frames from one node to another node over link.

→ This can be wired LAN/WAN or wireless LAN/WAN.

→ The data-link layer
- gets the datagram from network layer.
- encapsulates the datagram in a packet called a frame.
- Sends the frame to physical layer.

→ It supports all the standard and proprietary protocols.



3) Network Layer :-

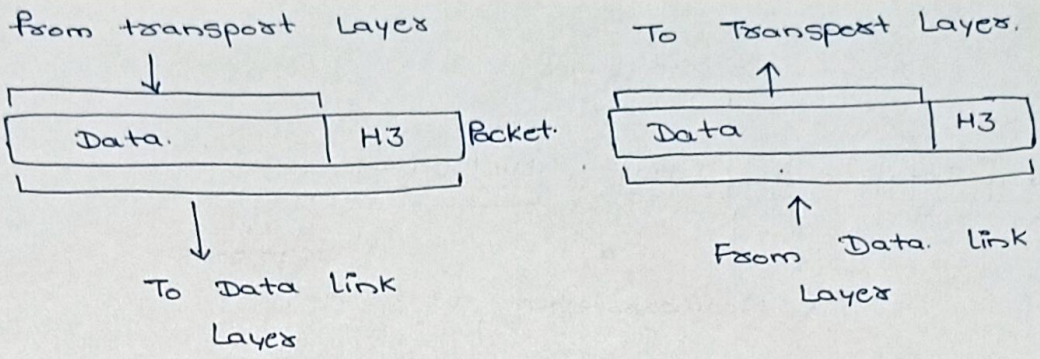
→ It is responsible for source to destination transmission of data.

→ Network Layer is responsible for routing the packet.

→ It is responsible for host to host communication.

→ The network layer includes unicast (one-to-one) and multicast (one-to-many) routing protocols.

→ TCP/IP model defines 4 protocols :-
1. IP (Internetworking Protocol)
2. ARP (Address Resolution Protocol)
3. ICMP (Internet Control Message Protocol)
4. IGMP (Internet Group Message Protocol)



4) Transport Layer :-

→ TL protocols are responsible for delivery of a message from a process to another process.

→ TL gets the message from the application Layer.

→ It encapsulates the message in a packet called a Segment & Sends the Segment to network Layer.

→ TCP/IP model defines 3 protocols:

* TCP (Transmission Control Protocol) :-

→ TCP is a reliable connection oriented.

→ A connection b/w Sender & receiver is established before the data can be transmitted.

→ TCP provides
  - Flow Control
  - Error Control
  - Congestion Control

* UDP (User Datagram Protocol) :-

→ It is an unreliable, connectionless protocol

→ It does not provide flow control, error or congestion control

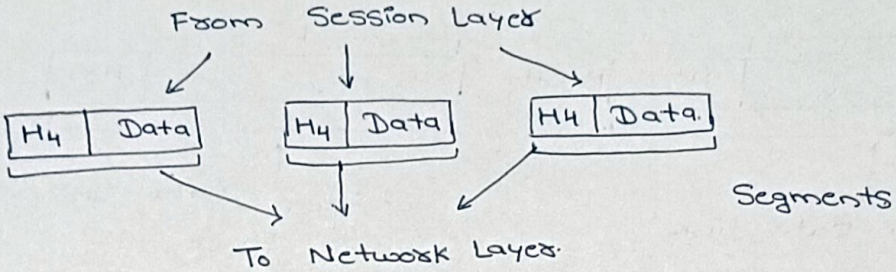
→ Each datagram is transported separately & independently.

→ It is suitable for application program that needs to send short messages & cannot afford the retransmissi

* SCTP (Stream Control Transmission protocol) :-

→ SCTP provides support for newer applications such as voice over the Internet.

→ It combines the best features of UDP and TCP.



5) Application Layer :-

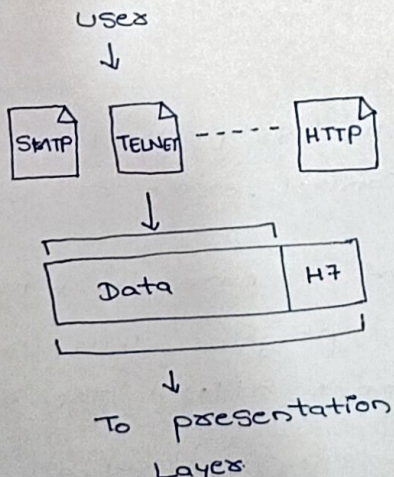
→ The two application layers exchange message between each other.

→ To communicate, a process sends a request to the other process and receives a response.

→ process-to-process communication is the duty of the application layer.

* TCP/IP model defines following protocols :-

① SMTP – for Email
② TELNET – for accessing a site remotely
③ FTP – for transforming files
④ DNS – to find IP address
⑤ SNMP – to manage internet at global level
⑥ HTTP – for accessing WWW



→ Transmission impairment occurs when the received signal is different from the transmitted signal.

→ In Analog Signals due to transmission impairment resulting received signal gets different amplitude or the shape.

→ In the case of digitally transmitted signals at the receiver side we get changes in bits (0's & 1's).

1) Attenuation :-

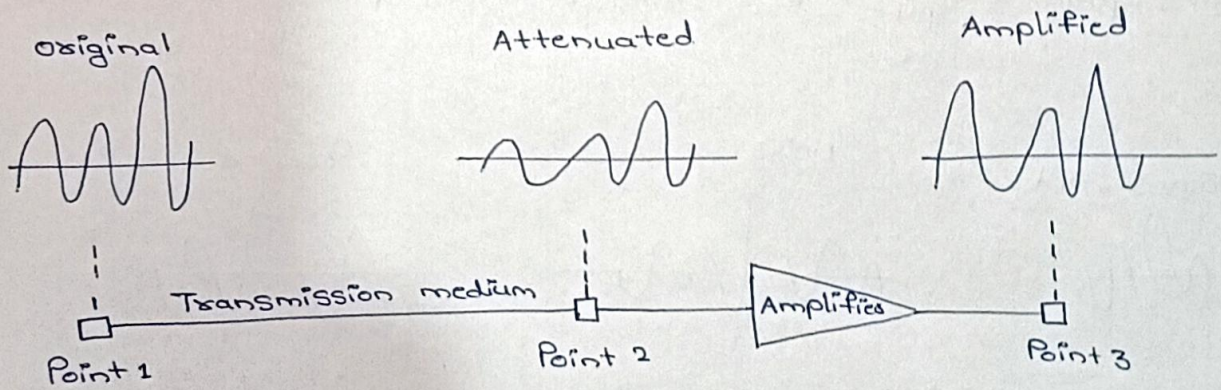
→ Means loss of energy.

→ When a signal travels through a medium it loses energy overcoming the resistance of the medium.

→ Amplifiers are used to compensate for this loss of energy by amplifying the signal.

→ $dB = 10 \log_{10} P_2/P_1$

P_1 - input Signal
P_2 - Output Signal



2) Distortion :-

→ Means that the Signal changes its form or shape.

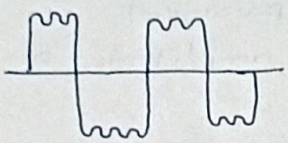
→ Distortion occurs in composite signals.

→ Each frequency component has its own propagation speed traveling through a medium.

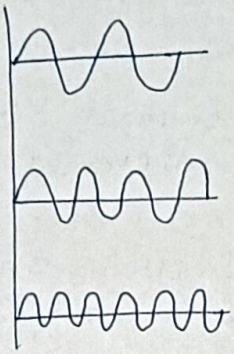
→ The different components therefore arrive with different delays at the receiver.

→ That means that the signals have different phases at the receiver than they did at the source.

Sender

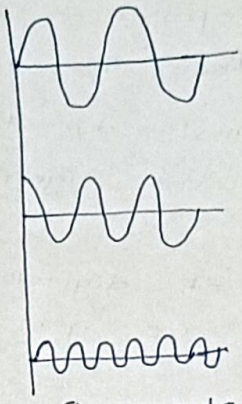


Composite Signal Sent

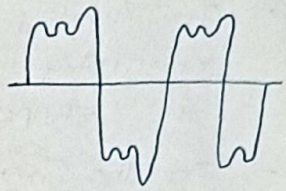


Components, in phase

Receiver



Components, out phase.



Composite Signal received

* Noise :-

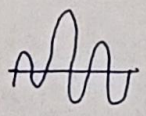
→ These are different types of noise

① Thermal :- Random noise of electrons in the wire. Creates an extra signal.

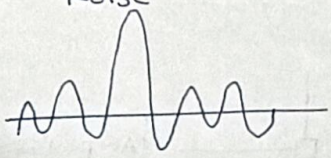
② Induced :- From motors and appliances, devices act. are. transmitter antenna and medium as receiving antenna.

③ Impulse :- Spikes that result from power lines, lightning, etc.

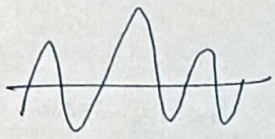
Transmitted



Noise



Received



* Digital to Digital Conversion :-

→ Digital to Digital encoding is the representation of digital information by a digital signal.

→ When binary 1s and 0's generated by the Computer are translated into a Sequence of voltage pulses that can be propagated over a wire, this process is known as digital-to-digital encoding.

* **Line Coding :-**

→ Line Coding is the process of converting digital data to digital signals.

→ Converting a string of 1's and 0's into a sequence of signals that denote the 1's and 0's

* **Data rate and Baud Rate :-**

→ $S = N/r$

→ $S_{ave} = c \times N \times 1/r$ bauds

where, N is data rate
S is signal rate
c is case factor
r is the ratio b/w data element & signal element

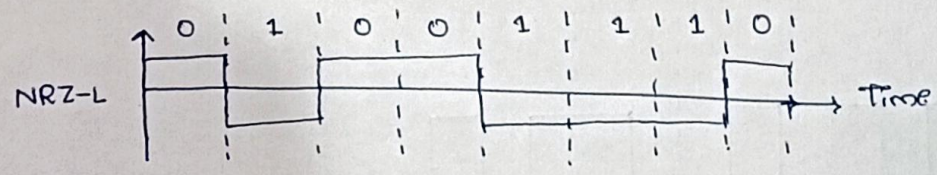
* **Polar :-**

→ NRZ, RZ, and biphase (Manchester and differential Manchester)

1) **Polar NRZ :-**

→ The voltage are on both sides of the time axis

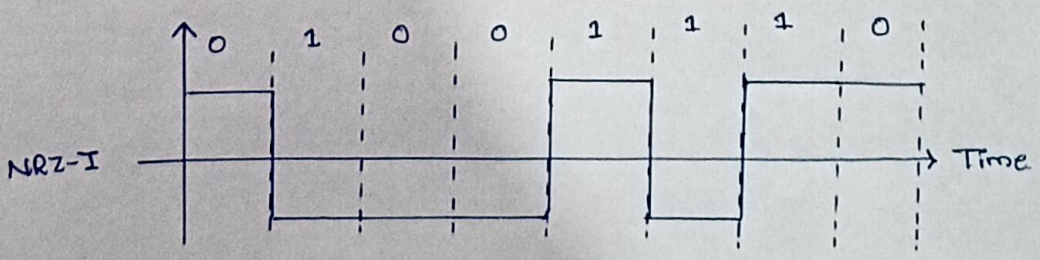
* **NRZ-L :-** positive voltage for one symbol & negative for other



* **NRZ-I :-** The signal changes (inverts) on a binary 1 and remains the same on a binary 0.

→ If the bit is 1, the signal flips.

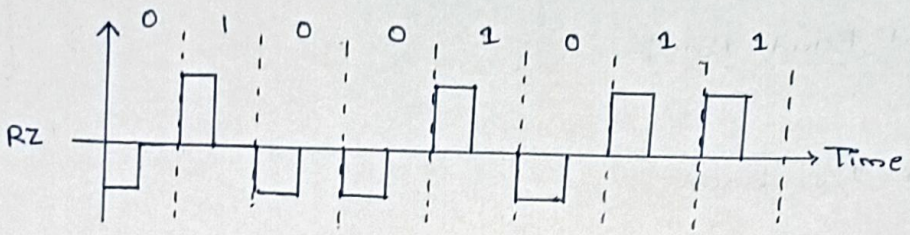
→ If the bit is 0, the signal stays same as previous bit.



2) Polar RZ :-

→ The return to zero (RZ) scheme uses three voltage values. +, 0, −

→ Each symbol has a transition in the middle. Signal goes to 0 in the middle of each bit.



3) Polar - Biphase :-

* Manchester : It consists of combination of NRZ-L & RZ schemes. 0 is ⊓ & 1 is ⊔

* Differential Manchester : Combination of NRZ-I & RZ schemes

